

Keplerian Orbital Equations

Radius	$r = \frac{p}{e \cos(\nu) + 1}$	Semi-Latus Rectum	$p = a \left(1 - e^2\right)$	Semi-Minor Axis	$b = a \sqrt{1 - e^2}$	Radius of Periapsis	$r_p = a \left(1 - e\right)$
Radius of Apoapsis	$r_a = a \left(e + 1\right)$	Velocity (elliptical)	$v = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}}$	Vis-Viva	$v = \sqrt{\mu} \sqrt{\frac{2}{r} - \frac{1}{a}}$	Specific Orbital Energy	$\varepsilon = -\frac{\mu}{2a} = -\frac{\mu}{r} + \frac{v^2}{2}$
Characteristic Energy	$C_3 = -\frac{\mu}{a}$	Velocity (circular)	$v_c = \frac{\sqrt{\mu}}{\sqrt{r}}$	Escape Velocity	$v_{esc} = \frac{\sqrt{2}\sqrt{\mu}}{\sqrt{r}}$	Hyperbolic Excess Velocity	$v_\infty = \sqrt{\mu} \sqrt{-\frac{1}{a}}$
Specific Angular Momentum	$h = \sqrt{\mu} \sqrt{p} = r v \cos(\gamma)$	Mean Anomaly (Elliptical)	$M = E - e \sin(E)$	Mean Motion	$n = \sqrt{\mu} \sqrt{\frac{1}{a^3}}$	Period	$T = \frac{2\pi \sqrt{a^3}}{\sqrt{\mu}}$
True Anomaly Rate	$\dot{\nu} = \frac{a^2 n \sqrt{1 - e^2}}{r^2}$	Radial Rate	$\dot{r} = \frac{\dot{\nu} e r \sin(\nu)}{e \cos(\nu) + 1}$	Argument of Latitude	$u = \nu + \omega$	Eccentric Anomaly	$E = \operatorname{atan}_2\left(\sqrt{1 - e^2} \sin(\nu), e + \cos(\nu)\right)$
True Anomaly	$\nu = \operatorname{atan}_2\left(\sqrt{1 - e^2} \sin(E), -e + \cos(E)\right)$	Flight Path Angle	$\gamma = \operatorname{atan}_2\left(e \sin(E), 1 - e^2\right)$	Semi-Latus Rectum (Parabolic)	$p = \frac{h^2}{\mu}$	Parabolic Anomaly	$B = \operatorname{atan}\left(\sin\left(\frac{\nu}{2}\right)\right)$
Flight Path Angle - Parabolic	$\gamma = \frac{\nu}{2}$	Hyperbolic Anomaly	$H = \operatorname{atan}_2\left(\sqrt{e^2 - 1} \sinh(\nu), e + \cosh(\nu)\right)$	Mean Anomaly (Hyperbolic)	$M = -H + e \sinh(H)$	Hyperbolic Turning Angle	$\epsilon = 2 \operatorname{asin}\left(\frac{1}{e}\right)$
Perifocal Radius	$\mathbf{r}_{PQW} = \begin{bmatrix} \frac{p \cos(\nu)}{e \cos(\nu) + 1} \\ \frac{p \sin(\nu)}{e \cos(\nu) + 1} \\ 0 \end{bmatrix}$	Perifocal Velocity	$\mathbf{v}_{PQW} = \begin{bmatrix} -\frac{\sqrt{\mu} \sin(\nu)}{\sqrt{p}} \\ \frac{\sqrt{\mu}(e + \cos(\nu))}{\sqrt{p}} \\ 0 \end{bmatrix}$	Perifocal to Inertial Rotation Matrix	$[\frac{IJK}{PQW}] = \begin{bmatrix} -\sin(\Omega) \sin(\omega) \cos(i) + \cos(\Omega) \cos(\omega) & -\sin(\Omega) \cos(\omega) \cos(i) - \sin(\omega) \cos(\Omega) & \sin(\Omega) \sin(i) \\ \sin(\Omega) \cos(\omega) + \sin(\omega) \cos(\Omega) \cos(i) & -\sin(\Omega) \sin(\omega) + \cos(\Omega) \cos(\omega) \cos(i) & -\sin(i) \cos(\Omega) \\ \sin(\omega) \sin(i) & \sin(i) \cos(\omega) & \cos(i) \end{bmatrix}$		
Radius Vector: Inertial	$r_{IJK} = [\frac{IJK}{PQW}] \mathbf{r}_{PQW}$	Velocity Vector: Inertial	$v_{IJK} = [\frac{IJK}{PQW}] \mathbf{v}_{PQW}$	Two Body Differential Equation	$\ddot{\hat{r}} = -\frac{\mu}{r^3} r_{IJK}$	Equinoctial Eccentricity Cosine Term	$f = e \cos(\Omega + \omega)$
Equinoctial Eccentricity Sine Term	$g = e \sin(\Omega + \omega)$	Equinoctial Inclination Cosine Term	$h = \cos(\Omega) \tan\left(\frac{i}{2}\right)$	Equinoctial Inclination Sine Term	$k = \sin(\Omega) \tan\left(\frac{i}{2}\right)$	Mean Longitude	$L = \Omega + \nu + \omega$
Delaunay L	$L_d = \sqrt{\mu} \sqrt{a}$	Delaunay H	$H_d = \sqrt{\mu} \sqrt{a(1 - e^2)} \cos(i)$				

Sources

Bates, Mueller, White — *Fundamentals of Astrodynamics*

- **Escape Velocity** — p. 35: Eq 1.9-2
- **Period** — p 33: Eq 1.7-9
- **Radius** — p. 20: Eq 1.5-4
- **Semi-Latus Rectum** — p. 24: Eq 1.5-6
- **Semi-Minor Axis** — p. 15: Eq 1.4
- **Velocity (circular)** — p. 34: Eq 1.8-2

Jacob Williams — *Modified Equinoctial Elements*

- **Equinoctial Eccentricity Cosine Term** — <https://degenerateconic.com/modified-equinoctial-elements.html>
- **Equinoctial Eccentricity Sine Term** — <https://degenerateconic.com/modified-equinoctial-elements.html>
- **Equinoctial Inclination Cosine Term** — <https://degenerateconic.com/modified-equinoctial-elements.html>
- **Equinoctial Inclination Sine Term** — <https://degenerateconic.com/modified-equinoctial-elements.html>
- **Mean Longitude** — <https://degenerateconic.com/modified-equinoctial-elements.html>

Vallado — *Fundamentals of Astrodynamics and Applications* — 4th Edition

- **Argument of Latitude** — 2-90 p. 102
- **Characteristic Energy** — p. 27
- **Delaunay L** — 2-102
- **Delaunay H** — 2-102
- **Eccentric Anomaly** — Simplified from other expressions
- **Flight Path Angle** — Simplified from other expressions

- **Flight Path Angle - Parabolic** — p. 2
- **Hyperbolic Anomaly** — Simplified from other expressions
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- **Hyperbolic Turning Angle** — 2-28
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- **Mean Anomaly (Hyperbolic)** — p. 2
- **Mean Motion** — p. 45: Eq 2-5
- **Parabolic Anomaly** — p. 2
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- **Perifocal to Inertial Rotation Matrix** — Algorithm 10, page 119
- **Perifocal Velocity** — 2-106
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- **Specific Orbital Energy** — p. 2
- **True Anomaly** — Simplified from other expressions
- **True Anomaly Rate** — p. 2
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- **Velocity (elliptical)** — p. 2
- **Velocity Vector: Inertial** — Algorithm 10, page 119
- **Vis-Viva** — p. 27, Eq. 1-22

Note on atan2: This sheet uses $\text{atan2}(y, x)$ (sine term/ y first, cosine term/ x second) to preserve quadrant.